S3.2 Functional groups: Classification of organic compounds

Types of formula

Learning outcomes

At the end of this section you should be able to:

- Identify different formulas and interconvert molecular, skeletal and structural formulas.
- Construct 3D models (real or virtual) of organic molecules.

What makes carbon such a unique element? The answer lies in the way in which carbon is able to form bonds with other carbon atoms. This process is known as catenation. Carbon can form straight-chain, branched-chain and ring structures with other carbon atoms (**Figure 1**). Other atoms such as oxygen, nitrogen and the halogens can bond to these chains that give the compounds their different properties.

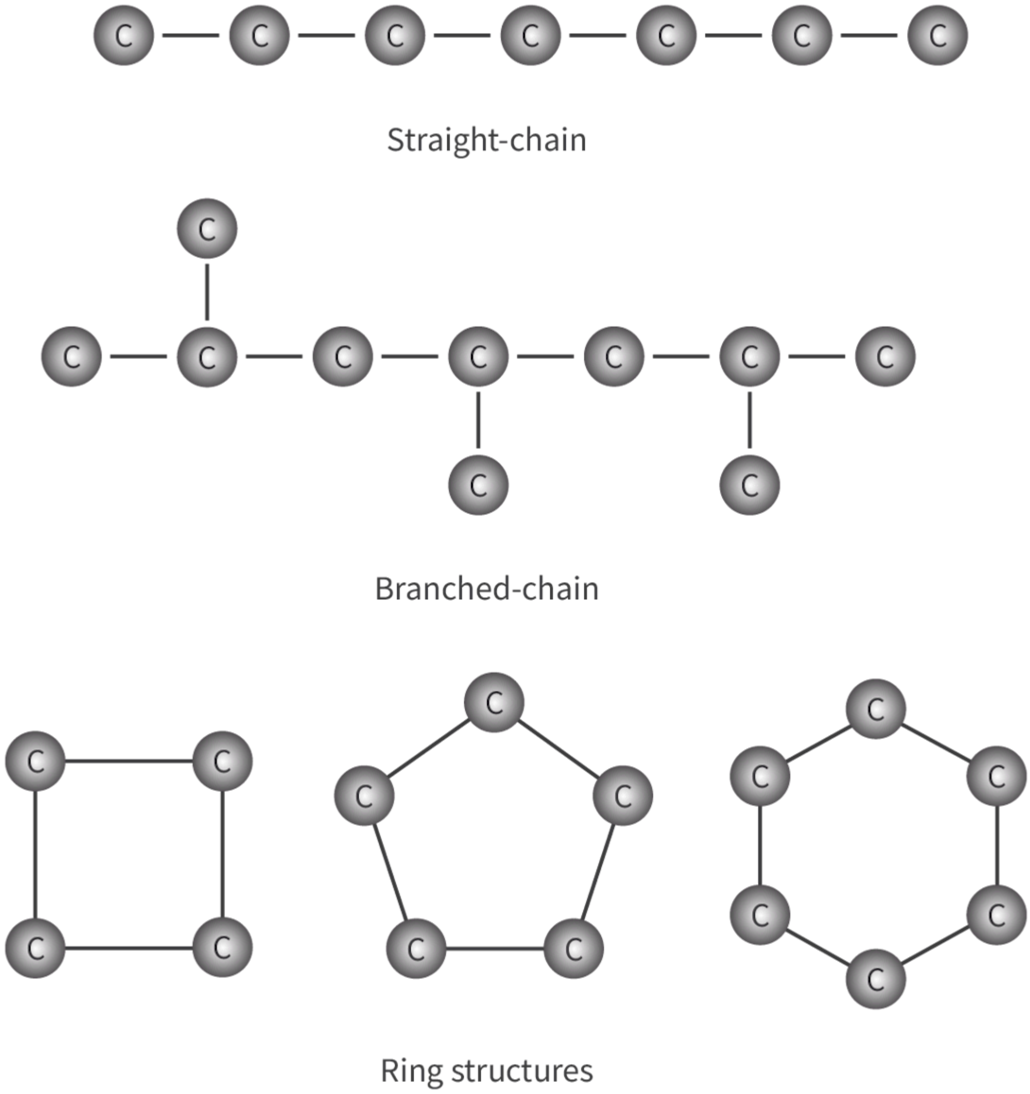


Figure 1. The catenation of carbon allows for the formation of millions of different organic compounds.

More information for figure 1

The image is a diagram illustrating the concept of carbon catenation through three types of structures: straight-chain, branched-chain, and ring structures.

1. **Straight-chain structure**: At the top, a series of carbon atoms are linked in a linear fashion, represented by circles labeled "C" connected by single lines, demonstrating the possible formation of a straight carbon chain.
2. **Branched-chain structure**: Below the straight-chain, a network shows carbon atoms branching. It starts with a linear arrangement but includes additional carbon atoms branching off the main line, highlighting potential branch points where additional atoms might attach.

3. **Ring structures:** At the bottom, three different geometric shapes made of carbon atoms are shown. These rings consist of a square, a pentagon, and a hexagon, all formed by carbon atoms connected in a closed loop. Each circle continues to be labeled "C" to denote carbon atoms, demonstrating how carbon atoms can also form circular structures.

Text labels accompany each grouping to identify them as "Straight-chain," "Branched-chain," and "Ring structures," elucidating the formation possibilities in organic compounds.

[Generated by AI]

Making connections

What is unique about carbon that enables it to form more compounds than the sum of all the other elements' compounds (see subtopic S2.2 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/the-big-picture-id-43109/))

What does a formula tell you about a compound? How many different types of formula are there? Are some types of formula easier to interpret than others? These are the types of question we will aim to answer in this section. So far in the DP chemistry course, you have learned about two different types of formula. One tells you the actual number of each type of element in a compound whereas the other tells you the lowest whole number ratio of atoms in a compound. Can you remember these two types of formula and their definitions? Try to name them both before checking your answer.

The two types of formula covered in section S1.4.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/empirical-formula-and-molecular-formula-id-43572/) are molecular and empirical formulas. The molecular formula tells us the actual number of each type of element in a compound and the empirical formula tells us the lowest whole number ratio of atoms in a compound. As you will learn later, different compounds can have the same molecular formulas but have different structures. Take, for example, the two compounds shown in **Figure 2**, butane and 2-methylpropane. This explains why other types of formula are required to show the difference between the structures of these types of compound.

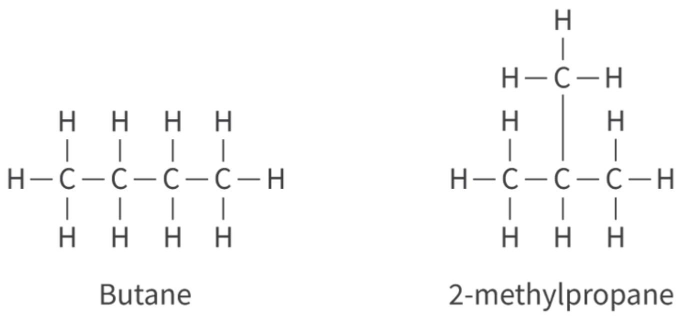


Figure 2. The full structural formulas of butane and 2-methylpropane.

Practical skills

Tool 1: Experimental techniques — Applying techniques

Throughout this section it might be helpful for you to have a molecular modelling kit available to build some of the molecules you will be learning about. Alternatively, there are a number of online tools available for you to view the molecules, such as molview (https://molview.org/).

Organic compounds can be represented by a range of different types of formula. Each type of formula gives a different amount of detail about the structure of the molecule in question. In this section, you will learn about these different types of formula used to represent organic compounds.

Making connections

In section S2.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/) you learned how to name inorganic compounds according to the IUPAC nomenclature rules. In this section, you will apply these rules to name organic compounds.

Nature of Science

Aspect: Models

What are the advantages and disadvantages of different depictions of an organic compound (e.g. structural formula, stereochemical formula, skeletal formula, 3D models)? As you read through this section, think about the advantages and disadvantages of each type of formula.

To learn about each type of formula, we will look at examples of organic compounds represented using each type. The first example we will look at is butane, a gas used for cooking. The molecular formula of butane is C_4H_{10}, which is the actual number of atoms in the compound. The empirical formula is C_2H_5, which is the lowest whole number ratio of atoms in the compounds.

Study skills

The order of the elements in a molecular formula is carbon, C, followed by hydrogen, H, and then the other elements in the compound in alphabetical order. For example, the organic compound methanamide is made up of carbon, hydrogen, nitrogen and oxygen atoms. Its molecular formula is written as CH_3NO.

The full structural formula of butane was shown in **Figure 2** (in the show/hide solution box above). This type of formula gives us the most information about the compound: it shows the number and each type of atom as well as the bonds between the atoms. You may notice that these are similar to the Lewis formulas that you learned how to draw in section S2.2.1a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/content-to-come-id-43106/), however, one important difference is that whereas Lewis structures must show all bonding and non-bonding electrons (lone pairs), it is common not to include the non-bonding electrons when drawing full structural formulas. What type of bonds do you think butane has between its atoms? Think about the answer before checking.

The type of bonds between the atoms in butane are single covalent bonds, represented by a single line. Recall that a single line represents a single covalent bond, a double line a double covalent bond and a triple line a triple covalent bond. All organic compounds that you will learn about in this section have covalent bonds between their atoms.

However useful the full structural formula is, it is more common to use the condensed structural formula to represent an organic compound. The condensed structural formula of butane is:

$$CH_3CH_2CH_2CH_3$$

Note that the order of the atoms is the same as the full structural formula but the bonds between the atoms are omitted, which allows the formula to be condensed.

Next, we have the skeletal formula, which only shows the carbon backbone of the molecule. The carbon atoms and hydrogen atoms of the backbone are omitted. The only type of atoms actually shown are those such as oxygen, nitrogen or hydrogen atoms that are part of the functional group of the molecule.

The skeletal formula of butane is shown in **Figure 3**. To the right of the skeletal formula is the arrangement of the atoms. Which type of structure do you think most accurately represents the molecule?

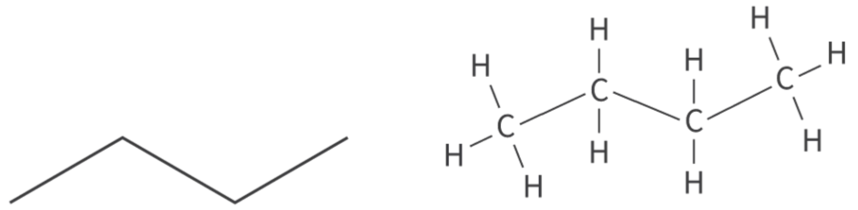


Figure 3. The skeletal formula of butane (left) and the arrangement of atoms (right).

More information for figure 3

The image consists of two parts: on the left, the skeletal formula of butane is depicted as a zigzag line representing the carbon backbone, without showing the hydrogen atoms. On the right, the arrangement of atoms in butane is shown with labeled carbon (C) and hydrogen (H) atoms. The structure on the right illustrates that each carbon atom is bonded to three hydrogen atoms and one other carbon atom, except for the terminal carbons, which are bonded to three hydrogen atoms. The carbon atoms are connected in a linear fashion, reflecting the straight-chain structure of butane.

[Generated by AI]

Figure 4 shows how to deduce the skeletal formula of a molecule from the full structural formula for the compound 4-methylpentan-1-ol.

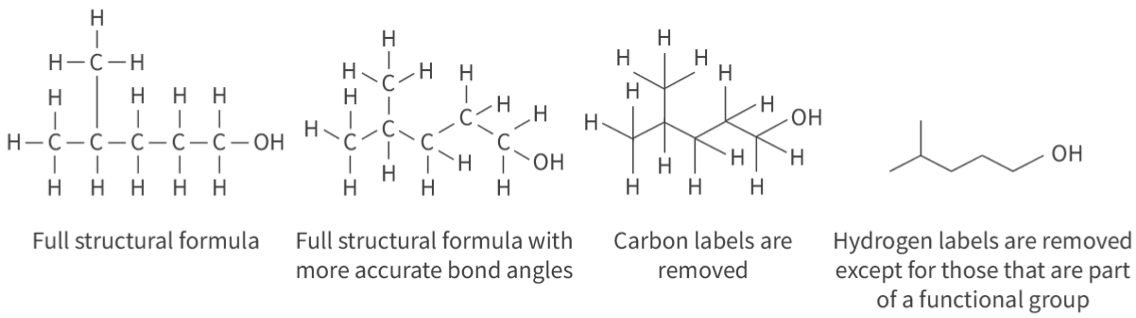


Figure 4. How to deduce a skeletal formula.

More information for figure 4

The image consists of four stages illustrating the process of deducing the skeletal formula of the compound 4-methylpentan-1-ol from its full structural formula. Starting from the left, the first diagram shows the full structural formula with each carbon and hydrogen atom explicitly labeled. The second diagram, labeled 'Full structural formula with more accurate bond angles,' adjusts the angles to reflect typical bond angles more accurately but maintains the hydrogen labels. In the third stage, labeled 'Carbon labels are removed,' only the hydrogen atoms are depicted along with the adjusted bond angles. Finally, in the fourth illustration, labeled 'Hydrogen labels are removed except for those that are part of a functional group,' hydrogen labels are removed except for those linked to the functional OH group, producing the skeletal formula: zigzag lines representing carbon connections and the OH group depicted on the far right.

[Generated by AI]

The skeletal formula is actually the most accurate in terms of the bond angles in the molecule. **Figure 5** shows the ball and stick model of butane. Use information from VSEPR theory in section S2.2.4 (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/vsepr-id-45129/) to predict the bond angle around each carbon atom (hint: count the number of electron domains around each carbon atom). Check your answer when you are ready.

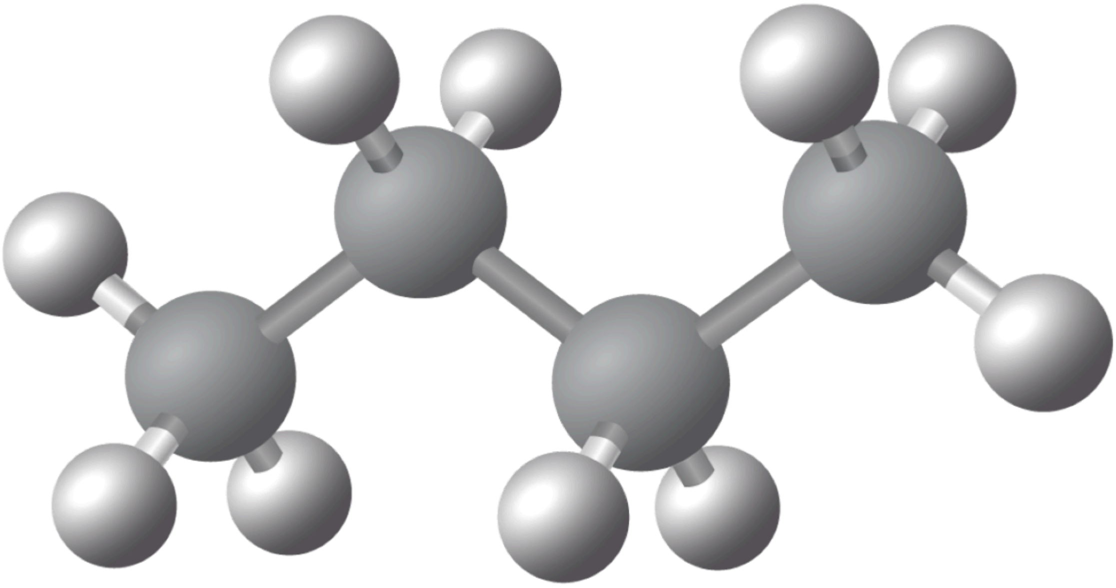


Figure 5. The ball and stick model of butane. What are the bond angles around each carbon atom?

More information for figure 5

The image shows a ball and stick model of a butane molecule. It illustrates a chain of four carbon atoms, each represented by a larger sphere, connected linearly by sticks indicating the covalent bonds. Each carbon atom is further bonded to hydrogen atoms, represented by smaller spheres. The structure demonstrates the spatial arrangement and bond angles within the molecule. This model visually explains the tetrahedral bonding around each carbon, crucial for understanding the spatial geometry and bond angles in butane based on VSEPR theory.

[Generated by AI]

There are four electron domains around each carbon atom (which are four bonding domains). So the bond angle, according to VSEPR theory, is 109.5°, and the electron domain geometry and molecular geometry are both tetrahedral.

The next molecule we will look at is ethanol, an organic compound that is commonly used in hand sanitiser. The full structural formula of ethanol is shown in **Figure 6**.

Kog

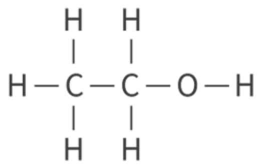


Figure 6. The full structural formula of ethanol.

More information for figure 6

The image displays the structural formula of ethanol, an organic compound. It consists of a central carbon (C) chain with two carbon atoms connected by a single bond. The first carbon atom is attached to three hydrogen (H) atoms, while the second carbon atom is attached to two hydrogen atoms and a hydroxyl (OH) group. The hydroxyl group is composed of an oxygen (O) atom bonded to a hydrogen atom. This structure shows the typical arrangement found in ethanol, providing a clear depiction of the molecular composition.

[Generated by AI]

This time, you will try to determine the different types of formula before checking your answers.

First, try to write the molecular formula and empirical formula for ethanol. When you have done so, check your answer.

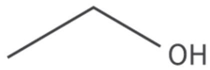
The molecular formula is C_2H_5OH (or C_2H_60), which is the same as the empirical formula.

Next, try to write the condensed structural formula for ethanol. Remember to remove the bonds between the atoms and group the atoms together.

The condensed structural formula is CH_3CH_2OH.

Finally try to sketch the skeletal formula for ethanol. Hint: the only atoms shown are the OH atoms that make up the functional group.

The skeletal formula is shown below. Don't worry if you weren't able to sketch it correctly. It will take practice before you can sketch this type of formula. Here, you can see that the carbon and hydrogen atoms that form the backbone of the molecule are not shown but the atoms that make up the functional group (the OH) are shown.



The ball and stick model of ethanol is shown in **Figure 7**.

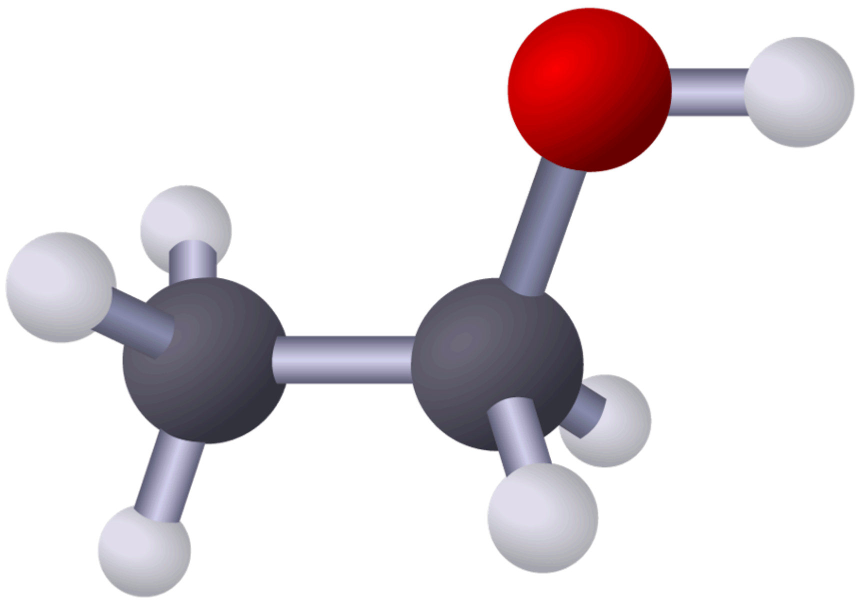


Figure 7. The ball and stick model of ethanol.

More information for figure 7

The image shows a ball and stick model representing the chemical structure of ethanol. The model is composed of interconnected spheres and rods, symbolizing atoms and chemical bonds, respectively. The ethanol molecule consists of two carbon atoms, six hydrogen atoms, and one oxygen atom. The carbon atoms are represented as two central dark spheres connected by a rod, forming the carbon backbone. The hydrogen atoms appear as smaller white spheres attached to the carbons, each connected by short rods. A red sphere, representing the oxygen atom, is bonded to one of the carbon atoms and has a white sphere (hydrogen) connected to it, illustrating the hydroxyl group. The model highlights the three-dimensional arrangement of atoms in ethanol.

[Generated by AI]

Worked example 1

From the full structural formula, determine the following.

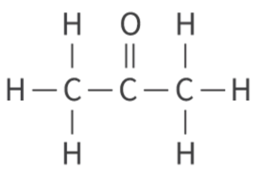


Figure 8. Full structure formula.

More information for figure 8

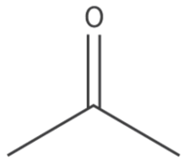
This image is a diagram showing the structural formula of a propanone molecule. At the center of the diagram, there is a carbon atom bonded to an oxygen atom with a double bond (known as a carbonyl group). On either side of the carbonyl group, there is a carbon atom bonded to three hydrogen atoms completing their four bonds typically required for carbon.

[Generated by AI]

1. The molecular and empirical formulas.
2. The condensed structural formula.
3. The skeletal formula.

1. The molecular formula is C_3H_6O which is the same as the empirical formula.
2. The condensed structural formula is CH_3COCH_3
3. The skeletal formula is shown here.

Kog



Worked example 2

From the full structural formula determine the following.

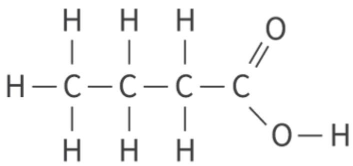


Figure 9. Full structural formula.

More information for figure 9

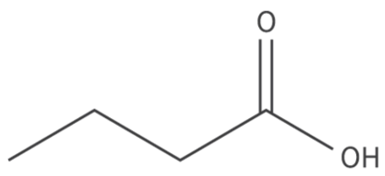
The image shows a full structural formula of a chemical compound. It features a linear arrangement of carbon (C) atoms bonded with hydrogen (H) and oxygen (O) atoms. The structure begins with a carbon atom on the left bonded to three hydrogen atoms, followed by two more carbon atoms, each bonded to two hydrogen atoms above and below. The chain continues ending with a fourth carbon atom double-bonded to an oxygen atom above and single-bonded to a hydroxyl group (OH) below. This structure likely represents a carboxylic acid functional group.

[Generated by AI]

1. The molecular and empirical formulas.
2. The condensed structural formula.

3. The skeletal formula.

1. The molecular formula is $C_4H_8O_2$
   The empirical formula is C_2H_4O
2. The condensed structural formula is $CH_3CH_2CH_2COOH$
3. The skeletal formula is shown here.



Stereochemical formula

A stereochemical formula is used to represent the 3D structure of an organic compound. This type of formula uses tapered bonds to represent the orientation of the bonds in a molecule.

- A wedge tapered bond represents a bond that is coming out of the plane of the paper (toward the viewer).
- A dashed tapered bond represents a bond going into the paper (away from the viewer).
- A straight line represents a bond in the same plane as the paper.

The ball and stick model of methane, together with the stereochemical formula, are shown in **Figure 10**.

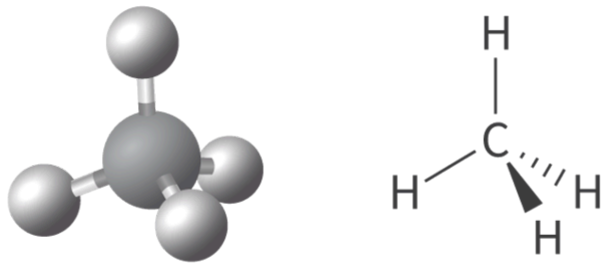


Figure 10. The ball and stick model of methane (left) and its stereochemical formula (right).

More information for figure 10

The image depicts two models of a methane molecule, side by side. On the left is a ball and stick model representing the molecule's three-dimensional structure. It shows one larger gray sphere representing a carbon atom, connected to four smaller gray spheres representing hydrogen atoms, each by a stick indicating the chemical bonds. On the right is the stereochemical formula of methane. It uses lines and symbols to depict the molecule's configuration. The central 'C' represents carbon, and lines extend outward to 'H' symbols for hydrogen, with one line depicted as a triangle and another as a dashed wedge to indicate stereochemistry.

[Generated by AI]

Higher level (HL)

Making connections

You are only expected to draw stereochemical formulas in certain sections: one of these is section S3.2.7a (https://app.kognity.com/study/app/preview-p/sid-426-cid-753301/book/stereoisomers-cistrans-isomerism-hl-id-45666/).

Nature of Science

Aspect: Models

Kog

Tool 2: Technology — Applying technology to collect data

Think about the advantages of using 3D models when learning about molecules that cannot be seen with the unaided eye.

In this activity, you will use 3D modelling software to model some organic compounds.

Activity

- **IB learner profile attribute:** Inquirer
- **Approaches to learning:** Communication skills — Using terminology, symbols and communication conventions consistently and correctly
- **Time required to complete activity:** 30 minutes
- **Activity type:** Individual activity

Click on the link to open molview (https://molview.org/?cid=91415364).

Download the worksheet.

Worksheet (https://d3vrb2m3yrmyfi.cloudfront.net/media/edusys_2/content_uploads/Chemistry

In molview, type the names of the organic compounds in the name bar at the top. For each compound in the table, write or sketch the different formulas. The first one has been done for you.